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68770 (SE-3)

Convert Infix $\rightarrow$ Post fix.

Infix:-

$$1. A + B * (C^D - E)^{(F + G * H)} - I$$

Steps	STACK	Output
A		A
+	+	A
B	+	AB
*	+*	AB
(	+*(	AB
C	+*(^	ABC
^	+*(^	ABC
D	+*(^	ABCD
-	+*(-	ABCD^
E	+*(-	ABCD^E
)	+*	ABCD^E-
^	+*^	ABCD^E-
(	+*^(	ABCD^E-F
F	+*^(	ABCD^E-F
+	+*^(+	ABCD^E-FG
G	+*^(+	ABCD^E-FG
*	+*^(+*	ABCD^E-FGH
H	+*^(+*	ABCD^E-FGH*
)	+*^	ABCD^E-FGH*+
-	-	ABCD^E-FGH*+^*+
I	-	ABCD^E-FGH*+^*+I-

2. $A + (B * C - (D / E ^ F) * G) * H$

Step	Stack	Output
A		A
+	+	A
(	+(	A
B	+(	AB
*	+(*	AB
C	+(*	ABC
-	+(-	ABC*
(	+(-	ABC*
D	+(- (	ABC*D
/	+(- (/	ABC*D
E	+(- (/	ABC*DE
^	+(- (/^	ABC*DE
F	+(- (/^	ABC*DEF
)	+(-	ABC*DEF^/
*	+(- *	ABC*DEF^/
G	+(- *	ABC*DEF^/G
)	+(- (/	ABCDEF^/G-
*	+	ABCDEF^/G-H
H	+	ABCDEF^/G-H*+

Convert Infix $\rightarrow$ Prefix

$\rightarrow A + (B * (C + (D * E)))$

Step: 1

Reverse:

$((E * D) + C) * B + A$

Step	STACK	Output.
(	(	
(	((	
E	((	E
*	((*	E
D	((*	ED
)	(	ED*
+	(+	ED*
C	(+	ED*C
)		ED*C+
*	*	ED*C+
B	*	ED*C+B
+	+	EDC+B
A	+	EDC+BA
End		EDC+BA+

Reverse Output: $+ A * B + C * DE$

2. $A * (B + C * (D - E)) + F / G$

Reverse:-

$$G / F + ((E - D) * C + B) * A$$

Step	Stack	Output
G		G
/	/	G
F	/	GF
+	+	GF/
(	+(	GF/
(	+((	GF/
E	+((	GF/E
-	+((-	GF/E
D	+((-	GF/ED
)	+(	GF/ED-
*	+(*	GF/ED-
C	+(*	GF/ED-C
+	+(+	GF/ED-C*
B	+(+	GF/ED-C*B
)	+	GF/ED-C*B+
*	+	GF/ED-C*B+
A	++	GF/ED-C*B+A

End

Reverse

$$+ * A + B * C - D E / F G$$

Evaluate the following

↳ $56 + 2 * 34 + -$

Step	Output
5	5
6	5, 6
+	11 (5+6)
2	11, 2
*	22 (11 * 2)
3	22, 3
4	22, 3, 4
+	22, 7 (3+4)
-	15 (22-7)

↳ $- * 45 + 23$

Step	Output
3	3
2	2, 3
+	5 (2+3)
5	5, 5
4	4, 5, 5
*	20 (4 * 5), 5
-	15 (20-5)

↳ $93 * 42 / -$

Step	Output
9	9
3	9, 3
*	27 (9 * 3)
4	27, 4
2	27, 4, 2
/	27, 2 (4/2)
-	25 (27-2)

↳ $+ * 234$

Step	Output
4	4
3	3, 4
2	2, 3, 4
*	6 (2 * 3), 4
+	10 (6 + 4)